

Week 1: Literature Review

*Determining the Role of Big Data Applications in Fraud Detection
and Financial Security in Financial Institutions of Ireland*

Student: Yashwanth

Course: Open Data Practice - Research Practicum

Supervisor: Dr. Prasanth Nayak

Document: Week 1 - Literature Review (2024-2026)

Papers: 15 Peer-Reviewed Papers

Date: September 2026

Each paper is reviewed under: What is the Paper | What Does It Do | Research Gap | Future Findings | Baseline Status

1. Introduction

This Week 1 literature review covers 15 peer-reviewed papers published between 2023 and 2026, directly relevant to the research on Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland. The review is structured across five thematic areas:

- (a) Big Data Applications in Fraud Detection (Papers 1-3)
- (b) ML Models for Credit Card Fraud Detection (Papers 4-6)
- (c) Class Imbalance & Resampling Techniques (Papers 7-9)
- (d) Financial Security & Regulatory Context (Papers 10-12)
- (e) Feature Engineering & Model Interpretability (Papers 13-15)

For each paper, four key questions are answered:

- | | |
|-----------------------|---|
| 1. What is the Paper? | - The scope and context of the study |
| 2. What Does It Do? | - The methodology, techniques and results |
| 3. Research Gap | - What the paper identified as missing in existing literature |
| 4. Future Findings | - Recommendations and future research directions |

Papers marked with [BASELINE] are designated as BASELINE PAPERS whose methodology is directly adopted or extended in the current study.

Section A: Big Data Applications in Fraud Detection (Papers 1-3)

Paper 1 [BASELINE] BASELINE PAPER

Big Data Analytics and Its Usage on Financial Fraud Detection in the USA

Author

URL: <https://www.researchgate.net/publication/393634924>

What is the Paper?:

This paper investigates how big data analytics technologies - including Hadoop, Spark, and real-time streaming - are deployed by financial institutions in the USA for detecting and preventing fraud. It reviews modern data pipeline architectures that process millions of transactions daily and evaluates ML classifiers embedded within these pipelines.

What Does It Do?:

The study benchmarks multiple big data frameworks (MapReduce, Spark MLlib, and Flink) for their throughput and latency when running fraud detection models. It finds that Spark-based pipelines with Random Forest classifiers achieve the best trade-off between detection accuracy (F1 = 0.89) and processing speed. The paper also demonstrates that big data platforms reduce false positive rates by 35% compared to legacy rule-based systems.

Research Gap Addressed:

Most existing studies evaluated big data fraud detection in isolation from operational deployment. There is limited research on how these frameworks perform in production environments with real-time regulatory reporting requirements, especially outside the USA (e.g., in European/Irish financial markets).

Future Findings / Recommendations:

The authors recommend extending the study to European banking contexts where GDPR and PSD2 regulations impose additional constraints. They also suggest incorporating explainable AI (XAI) methods so fraud decisions can be audited by regulators.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 2

Leveraging Financial Data Analytics for Business Growth, Fraud Prevention, and Risk Mitigation in Markets

Author

URL: <https://www.researchgate.net/publication/390118650>

What is the Paper?:

This paper examines the broader role of financial data analytics - encompassing big data, predictive modelling, and anomaly detection - in fraud prevention and risk mitigation across emerging and developed financial markets.

What Does It Do?:

The study develops a conceptual framework that maps data analytics capabilities to specific fraud prevention outcomes. It reviews case studies from banking, insurance, and fintech sectors, demonstrating that institutions using advanced analytics detect fraud 2.5x faster than those relying on manual review processes. The paper quantifies that data-driven fraud prevention reduces financial losses by 40-60%.

Research Gap Addressed:

The paper identifies that while large multinational banks have adopted analytics, smaller regional and national

institutions - particularly in Ireland and other EU member states - lag behind due to resource constraints and regulatory uncertainty about AI-driven decision-making.

Future Findings / Recommendations:

Future research should focus on developing cost-effective, scalable analytics frameworks that smaller financial institutions can adopt, along with regulatory guidelines for AI-based fraud detection in the EU.

Paper 3

Using Data Analytics in Financial Statement Fraud Detection and Prevention: A Systematic Review of Methods, Challenges, and Future Directions

Author

URL: <https://www.mdpi.com/1911-8074/18/11/598>

What is the Paper?:

A comprehensive systematic literature review examining the current state of data analytics applications in financial fraud detection, covering both transactional fraud and statement fraud. The review synthesises findings from 80+ studies published between 2020-2025.

What Does It Do?:

The review categorises fraud detection methods into supervised (classification), unsupervised (clustering/anomaly), and hybrid approaches. It finds that ensemble methods - particularly Random Forest and Gradient Boosting - consistently outperform single classifiers across datasets. The paper provides a meta-analysis showing median F1 scores of 0.82 for ensemble methods vs 0.65 for single classifiers in fraud detection tasks.

Research Gap Addressed:

Most reviewed studies use the same few benchmark datasets (Kaggle Credit Card, IEEE-CIS), leaving a gap in validation on real-world proprietary banking data. There is also insufficient research on how these methods adapt to evolving fraud patterns (concept drift).

Future Findings / Recommendations:

The authors recommend longitudinal studies that track model performance over time as fraud patterns evolve, and call for public-private partnerships to create more diverse, representative fraud datasets.

Section B: ML Models for Credit Card Fraud Detection (Papers 4-6)

Paper 4 [BASELINE] BASELINE PAPER

Fraud Detection: A Hybrid Approach with Logistic Regression, Decision Tree, and Random Forest

URL: https://assets.cureusjournals.com/artifacts/upload/original_article/pdf/2350/20250210-51856-...

What is the Paper?:

This paper directly evaluates the three ML algorithms - Logistic Regression, Decision Tree, and Random Forest - for credit card fraud detection, making it the primary baseline for the current study. It uses the Kaggle Credit Card Fraud Detection dataset and compares model performance under class imbalance.

What Does It Do?:

The paper implements all three models with balanced class weights and evaluates them using accuracy, precision, recall, F1-score, and ROC AUC. Random Forest achieved the highest F1-score (0.85) and precision (0.94), while Logistic Regression achieved the highest recall (0.92). Decision Tree showed the best interpretability but lowest overall performance. The authors used stratified train-test split (80/20).

Research Gap Addressed:

The study did not apply advanced resampling techniques (SMOTE, ADASYN) to address class imbalance - it relied solely on `class_weight='balanced'`. It also did not consider the Irish/European regulatory context (GDPR implications) for deploying such models in production.

Future Findings / Recommendations:

The authors recommend combining oversampling with ensemble methods and evaluating model fairness metrics. They also suggest testing on larger, more diverse datasets beyond the standard Kaggle benchmark.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 5 [BASELINE] BASELINE PAPER

Credit Card Fraud Detection Using KNN, Random Forest and Logistic Regression Algorithms: A Comparative Analysis

URL: <https://www.ijcs.net/ijcs/index.php/ijcs/article/view/3707>

What is the Paper?:

This paper provides a comprehensive comparative analysis of three supervised ML algorithms (KNN, Random Forest, Logistic Regression) for credit card fraud detection, serving as another key baseline for model comparison methodology.

What Does It Do?:

Using the Kaggle Credit Card Fraud Detection dataset, the study compares KNN (k=5), Random Forest (100 trees), and Logistic Regression across multiple metrics. Random Forest achieved the best overall performance with AUC of 0.98 and F1-score of 0.86. The paper includes detailed analysis of confusion matrices, ROC curves, and precision-recall curves. It also analyses the computational cost of each algorithm.

Research Gap Addressed:

The study used only the standard Kaggle dataset without testing on any real-world banking data. Feature engineering was limited to PCA-transformed features already in the dataset. No exploration of how model performance degrades

when fraud patterns change over time.

Future Findings / Recommendations:

The authors suggest incorporating deep learning approaches (LSTM, Autoencoders) for temporal fraud pattern detection and testing with streaming data frameworks for real-time deployment.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 6

AI and ML Approaches for Credit Card Fraud Detection: A Comparative Study of Logistic Regression and Decision Tree Techniques

URL: <https://ieeexplore.ieee.org/abstract/document/10915092/>

What is the Paper?:

This IEEE conference paper focuses on comparing Logistic Regression and Decision Tree algorithms for credit card fraud detection, providing additional validation for the model comparison approach used in the current study.

What Does It Do?:

The paper evaluates Logistic Regression and Decision Tree on the Kaggle Credit Card dataset using 70/30 stratified splits. Logistic Regression achieved 97.4% accuracy with recall of 0.89, while Decision Tree achieved 99.2% accuracy but lower recall of 0.78. The paper highlights that accuracy is a misleading metric for imbalanced datasets and advocates for F1-score and PR-AUC as primary evaluation metrics.

Research Gap Addressed:

The study only compared two algorithms and did not include ensemble methods like Random Forest. It also lacked feature importance analysis, which is critical for understanding why certain transactions are flagged as fraudulent.

Future Findings / Recommendations:

The author recommends extending the comparison to include ensemble methods and applying SHAP values for model interpretability in regulatory contexts.

Author

Section C: Class Imbalance & Resampling Techniques (Papers 7-9)

Paper 7 [BASELINE] BASELINE PAPER

Comparative Analysis of Machine Learning Models for Credit Card Fraud Detection Using SMOTE for Class Imbalance

URL: <https://www.researchgate.net/publication/393024754>

What is the Paper?:

This paper directly addresses the class imbalance problem in credit card fraud detection using SMOTE (Synthetic Minority Over-Sampling Technique), which is the key challenge in the current study's dataset (0.173% fraud rate).

What Does It Do?:

The study applies SMOTE to the Kaggle Credit Card dataset before training Logistic Regression, Decision Tree, Random Forest, and SVM. Without SMOTE, Random Forest achieved F1=0.82; with SMOTE, F1 improved to 0.91. The paper also compares SMOTE with random undersampling and random oversampling, finding SMOTE achieves the best balance. Precision dropped slightly (from 0.93 to 0.87) but recall improved significantly (from 0.74 to 0.95).

Research Gap Addressed:

The paper used basic SMOTE without advanced variants (Borderline-SMOTE, SMOTE-ENN, ADASYN). It did not evaluate whether SMOTE introduces overfitting on the synthetic fraud samples. Cross-validation was not used to validate the improvements.

Future Findings / Recommendations:

The authors recommend testing SMOTE-ENN (combined oversampling + cleaning) and using k-fold cross-validation to ensure improvements are robust.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 8

Enhancing the Random Forest Model via Synthetic Minority Oversampling Technique for Credit-Card Fraud Detection

URL: <https://dl.futuretechsci.org/id/eprint/33/>

What is the Paper?:

This paper focuses specifically on improving Random Forest performance for fraud detection by applying SMOTE, which is directly relevant to the current study where Random Forest is identified as the best-performing model.

What Does It Do?:

The study demonstrates that applying SMOTE before Random Forest training on the Kaggle Credit Card dataset improves recall from 0.76 to 0.92 while maintaining precision above 0.85. The authors use a 70/30 stratified split and evaluate using F1-score, ROC-AUC, and Matthews Correlation Coefficient (MCC). They also compare SMOTE with ADASYN and find SMOTE provides more consistent results.

Research Gap Addressed:

The paper only evaluated Random Forest and did not compare with other classifiers to show whether SMOTE benefits are model-specific. Feature selection techniques were not explored alongside SMOTE.

Future Findings / Recommendations:

Future work should combine SMOTE with feature selection and evaluate on streaming/real-time data. The authors also recommend testing with different Random Forest hyperparameters (tree depth, number of estimators).

Paper 9

A Systematic Review of Machine Learning in Credit Card Fraud Detection Under Original Class Imbalance

Author

URL: <https://www.mdpi.com/2073-431X/14/10/437>

What is the Paper?:

A comprehensive systematic review examining how different ML algorithms perform specifically under the original class imbalance conditions found in credit card fraud datasets, without applying any resampling.

What Does It Do?:

The review analyses 50+ studies and finds that under original class imbalance (without SMOTE/resampling), ensemble methods achieve median F1 of 0.78 while single classifiers achieve only 0.45. Random Forest and XGBoost are identified as the most robust algorithms under severe imbalance. The paper also analyses how evaluation metric choice (accuracy vs F1 vs PR-AUC) affects conclusions about model quality.

Research Gap Addressed:

The review notes that most studies report only accuracy (which is misleading at 99%+ for imbalanced data) and do not report PR-AUC, which is the most informative metric for rare-event detection. Very few studies test on imbalance ratios worse than 1:100.

Future Findings / Recommendations:

The authors recommend that all future fraud detection studies report a standardised set of metrics (precision, recall, F1, PR-AUC, ROC-AUC) and specify the exact imbalance ratio of their dataset.

Section D: Financial Security & Regulatory Context (Papers 10-12)

Paper 10

Next Generation Financial Security: Leveraging AI for Fraud Detection, Compliance and Adaptive Risk Management

URL: <https://www.researchgate.net/publication/394087142>

What is the Paper?:

This paper examines how AI-driven fraud detection systems can be integrated with compliance and risk management frameworks in financial institutions, addressing the intersection of technology and regulation.

What Does It Do?:

The paper develops a multi-layered framework that combines ML-based fraud detection with automated compliance checking against regulatory requirements (KYC, AML, GDPR). Testing on synthetic banking data, the framework detected 94% of fraudulent transactions while generating compliance-ready audit trails. The paper demonstrates that integrating compliance reduces deployment barriers in regulated industries.

Research Gap Addressed:

The framework was tested only on synthetic data and not validated with real banking institutions. The regulatory analysis focused on US/UK regulations and did not address EU-specific regulations (PSD2, DORA) or Irish Central Bank requirements.

Future Findings / Recommendations:

The authors recommend pilot testing with real financial institutions and adapting the framework for EU/Irish regulatory requirements, including GDPR's right to explanation for automated decisions.

Paper 11

Artificial Intelligence-Driven Fraud Detection in FinTech: Strengthening Cybersecurity Against Digital Financial Scams

URL: <https://insightfuljournals.com/index.php/JBII/article/view/116>

What is the Paper?:

This paper investigates AI-driven approaches for fraud detection specifically in FinTech environments, where transaction volumes and fraud patterns differ from traditional banking.

What Does It Do?:

The study evaluates Random Forest, XGBoost, and LSTM networks for detecting digital payment fraud in FinTech platforms. Random Forest achieved 99.3% accuracy and F1=0.84 on a FinTech transaction dataset. The paper also implements real-time scoring with average latency under 50ms, demonstrating production-readiness. It analyses how FinTech-specific features (device fingerprint, geo-location, session behaviour) improve detection.

Research Gap Addressed:

The paper focused on FinTech-specific fraud patterns and did not compare with traditional credit card fraud detection. The dataset used was proprietary and not publicly available for reproduction.

Future Findings / Recommendations:

The authors recommend developing open benchmark datasets for FinTech fraud detection and exploring transfer

learning to adapt models trained on one fraud domain to another.

Paper 12

Impact of Data Analytics on Fraud Prevention and Public Trust in the United States

Author

URL: <https://www.researchgate.net/publication/402660479>

What is the Paper?:

This paper explores how data analytics implementations in fraud prevention affect public trust in financial institutions, providing the socio-economic context that supports the current study's research objectives.

What Does It Do?:

Through a mixed-methods study combining quantitative analysis of fraud detection metrics with qualitative surveys of 500+ banking customers, the paper finds that institutions implementing advanced analytics see a 28% increase in customer trust scores. It also quantifies that for every EUR1 invested in analytics-based fraud prevention, institutions save EUR4.70 in avoided fraud losses.

Research Gap Addressed:

The study focused exclusively on the US market. European/Irish financial institutions operate under different regulatory frameworks (GDPR, Central Bank of Ireland oversight) that may affect how analytics can be deployed and how public trust is influenced.

Future Findings / Recommendations:

The authors recommend replicating the study in European markets, particularly Ireland where EUR160 million in payment fraud was reported in 2024 (Central Bank of Ireland, 2025).

Section E: Feature Engineering & Model Interpretability (Papers 13-15)

Paper 13

Optimizing Machine Learning Classifiers for Credit Card Fraud Detection on Highly Imbalanced Datasets Using PCA and SMOTE Techniques

URL: <https://vfast.org/journals/index.php/VTCS/article/view/1921>

What is the Paper?:

This paper investigates the combined effect of PCA-based feature transformation and SMOTE on ML classifier performance for credit card fraud detection, which is directly relevant since the Kaggle dataset uses PCA-transformed features V1-V28.

What Does It Do?:

The study demonstrates that combining PCA (for dimensionality reduction) with SMOTE (for rebalancing) before applying ML classifiers (LR, DT, RF, SVM) yields the best results. On the Kaggle Credit Card dataset, the PCA+SMOTE+Random Forest pipeline achieved F1=0.93, ROC-AUC=0.99, and precision=0.91. The paper confirms that PCA-transformed features retain sufficient discriminative power for fraud detection.

Research Gap Addressed:

The study did not explore alternative feature engineering approaches beyond PCA. It also did not analyse which specific PCA components (V1-V28) contribute most to fraud detection, limiting interpretability.

Future Findings / Recommendations:

Future work should apply feature importance analysis (SHAP, permutation importance) to PCA components and explore whether original features (before PCA) would yield better interpretability for regulatory explanations.

Paper 14

Credit Card Fraud Detection: Machine Learning and Deep Learning Advances, Challenges, and Future Directions

URL: https://www.itm-conferences.org/articles/itmconf/pdf/2025/09/itmconf_cseit2025_02023.pdf

What is the Paper?:

A comprehensive review paper covering both traditional ML and deep learning approaches for credit card fraud detection, mapping the evolution of the field and identifying current best practices.

What Does It Do?:

The review categorises approaches into (1) traditional ML (LR, DT, RF, SVM), (2) ensemble methods (XGBoost, stacking), and (3) deep learning (LSTM, CNN, Autoencoder). It finds that while deep learning achieves marginally better performance (F1 improvement of 2-5%), traditional ML methods - especially Random Forest - remain competitive and are more interpretable. The paper argues that for production deployment, interpretability and computational efficiency favour traditional ML.

Research Gap Addressed:

The review identifies that most studies do not address real-time deployment challenges, concept drift, or the computational cost of model retraining. Very few studies have been validated in European banking contexts with GDPR compliance.

Future Findings / Recommendations:

The author recommends a shift from pure performance optimization to holistic evaluation including interpretability, latency, and regulatory compliance. Hybrid approaches combining ML with rule-based systems are identified as a promising direction.

Paper 15

A Supervised Machine Learning Algorithm for Detecting and Predicting Fraud in Credit Card Transactions

Author

URL: <https://doi.org/10.1016/j.dajour.2023.100163>

What is the Paper?:

This foundational paper proposes a supervised ML framework for credit card fraud detection, comparing multiple classifiers and establishing baseline performance metrics widely cited in subsequent research.

What Does It Do?:

The paper evaluates Decision Tree, Random Forest, Logistic Regression, Naive Bayes, and SVM on the Kaggle Credit Card dataset. Random Forest achieved the highest accuracy (99.96%) and F1-score (0.86). The study also implements feature importance analysis, identifying V14, V12, V10, and V17 as the top predictive features - consistent with the current study's findings. Stratified k-fold cross-validation was used.

Research Gap Addressed:

The paper did not apply any resampling techniques to address the 577:1 class imbalance. It focused primarily on accuracy, which is misleading for highly imbalanced datasets. No analysis of computational cost or model deployment considerations was included.

Future Findings / Recommendations:

The authors recommend applying SMOTE and other resampling methods, exploring deep learning alternatives, and validating results on non-PCA datasets with interpretable features.

2. Baseline Papers & Research Gap Summary

Chosen Baseline Papers:

Paper 1: Big Data Analytics and Its Usage on Financial Fraud Detection in the USA

Jamil, M.H., Hossen, A., Talukder, S.I., Arafat, Y. et al. (2025) - Advances in Machine Learning & Artificial Intelligence

Paper 4: Fraud Detection: A Hybrid Approach with Logistic Regression, Decision Tree, and Rando...

Salunke, Y., Phalke, S., Madavi, M. et al. (2025) - Cureus Journal of Computer Science

Paper 5: Credit Card Fraud Detection Using KNN, Random Forest and Logistic Regression Algorith...

Saeed, V.A., Abdulazeez, A.M. (2024) - The Indonesian Journal of Computer Science

Paper 7: Comparative Analysis of Machine Learning Models for Credit Card Fraud Detection Using...

Andrade-Arenas, L., Yactayo-Arias, C. (2025) - International Journal of Advanced Computer Science and Applications (IJACSA)

Key Research Gaps Identified Across All Papers:

1. Limited Irish/European context: Most studies focus on US/Asian markets. Very few validate ML fraud detection models within the Irish financial regulatory framework (GDPR, Central Bank of Ireland, PSD2).
2. No combined evaluation of LR, DT, RF with SMOTE: Baseline papers (4, 5) compare models under class imbalance using only `class_weight='balanced'` but do not apply SMOTE or other resampling techniques alongside all three algorithms.
3. Class imbalance inadequately addressed: The Kaggle Credit Card dataset has a 577:1 imbalance ratio. Most studies report accuracy (99%+) as primary metric, which is misleading. Few studies use PR-AUC, which is the most appropriate metric for rare-event detection.
4. Missing feature importance analysis for PCA features: Studies using the Kaggle dataset rely on PCA-transformed features (V1-V28) but do not systematically analyse which components are most predictive, limiting model interpretability.
5. No big data scalability assessment: Papers on ML fraud detection use small-scale batch processing. There is no evaluation of how these models would perform when integrated into big data platforms (Hadoop/Spark) for real-time detection in Irish banks.

How the Current Study Addresses These Gaps:

1. Irish financial context: This study frames the analysis within Irish financial institutions, referencing the Central Bank of Ireland's EUR160 million fraud loss report (2024) and GDPR/Irish Data Protection Act 2018 compliance considerations.
2. Comprehensive model comparison: Implements and compares Logistic Regression, Decision Tree, and Random Forest on the Kaggle Credit Card dataset using balanced class weights, with detailed evaluation across accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.
3. Proper imbalance handling: Uses `class_weight='balanced'` in all models and evaluates performance using F1-score and PR-AUC as primary metrics (not misleading accuracy). The 577:1 imbalance ratio is explicitly analysed and discussed.
4. Feature importance analysis: Conducts systematic feature importance analysis across all three models - tree-based importances for DT/RF and coefficient analysis for LR - identifying V14 as the strongest fraud predictor, consistent with

baseline findings.

5. Big Data recommendations: Provides actionable recommendations for implementing these ML models within big data frameworks for Irish financial institutions, addressing scalability and regulatory compliance.